

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

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Project Report

Synchronous Motor Rewinding

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1. Abstract

The project is a thorough investigation and optimization effort targeted at rewinding methods of synchronous motor, an essential element in several industrial and commercial applications. Through creative rewinding techniques, this initiative seeks to increase the dependability, effectiveness, and sustainability of these motors. We looked at how different winding arrangements, insulation types, and winding procedures affect the efficiency of motors. This study's main goal was to look at the rewinding process, how it affects motor efficiency and whether it can meet or exceed original requirements. The project approach included a detailed experimentation of coil design, insulating materials, and methods of synchronous motor winding. To assess the performance of the rewound motor under various circumstances, practical experiments were carried out. The study also looked at the advantages of motor rewinding over motor replacement in terms of the economy and the environment. By presenting our findings and insights from this project, we contribute valuable knowledge to the fields of electrical engineering and industrial maintenance. Our work not only benefits the industry by improving the performance and sustainability of synchronous motors but also aligns with broader initiatives for energy efficiency and reduced environmental impact. We invite you to explore the details of our research, which has the potential to revolutionize the way synchronous motors are maintained and contribute to a more sustainable future.

2. Introduction:

The **Synchronous Motor** is such a motor in which the **speed** of rotation of the rotor is identical to the synchronous speed, synchronous speed is the rotational speed of the magnetic field provided at the stator of the motor. The stator of the motor is electromagnetic and produces field when current is provided at the stator the rotating speed of the field is called synchronous speed. The rotor of a synchronous motor behaves like permanent magnetic or electromagnetic and its field interacts with the stator field then it rotates at a time with the rotation of the stator field. Synchronous motor is also known as the double **feed** because its rotor and stator are connected with separate input supplies.

Construction:

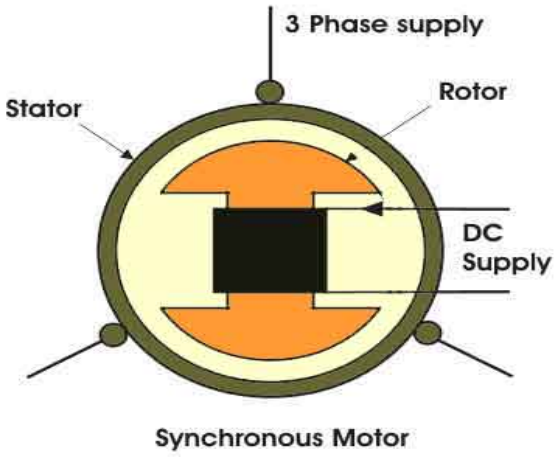


Figure 1: Construction of a synchronous motor

The stator and rotor are the main parts while frame is the cover. Stator and rotor both make up the electric and magnetic circuitry of Synchronous motor.

Stator Of Synchronous Motor: As the name suggests, a stator is the stationary part of motor. It is the outer frame which is cylindrical in shape. This cylindrical frame is laminated and has slots to carry winding circuitry. The stator of this motor is supplied with 3 AC power. The winding circuit of stator is called stator winding.

Rotor Of Synchronous Motor: Rotor is the rotating part, which rotates exactly at the same speed as the stator magnetic field. It is also cylindrical in shape, and it is the shaft of the motor from where output load is connected. It rotates in the stator frame which is separated by an air gap. There are two types of rotors. They are:

Permanent magnet rotor: This type of rotor has permanent magnets embedded in its surface. When the stator windings are energized, the magnetic field produced by the stator interacts with the magnets on the rotor, causing the rotor to rotate.

Excited rotor: A winding connected to a DC supply is located on the surface of this kind of rotor. The rotor creates a magnetic field when the DC source is powered. The magnetic field generated by the stator interacts with the magnetic field on the rotor when the stator windings are activated, spinning the rotor.

The rotor of a synchronous motor is excited by a DC source. There are two common approaches to supplying this dc power:

- Supply the dc power from an external dc source to the rotor by means of slip rings and brushes.
- Supply the dc power from a special dc power source mounted directly on the shaft of the synchronous generator.

The frame of the motor protects and covers the whole assembly. It can either be built in a vertical or horizontal shape and the type of covering depends on its area of use. That was all about construction of a synchronous motor.

Working of synchrounous motor:

The operation of a synchronous motor is basically that the rotor tries to follow the rotating magnetic field of stator and rotates at a speed approaching it. This motor is a doubly excited machine. The rotor winding is excited by a DC source while the stator winding is excited through an AC source. A 3-phase rotating magnetic field is produced by stator winding due to 3 phase AC. And a constant magnetic field is produced by rotor winding.

At some instant of rotation, the poles of two magnetic fields may attract each other while at some other instant, they repel each other. As a result, during one electrical cycle, the torque direction becomes clockwise (in half cycle) and counterclockwise (in other half cycle). The rotor will not start rotating due to these interactions due to its inertia. So, an external mean will compensate and provide initial rotation to the rotor. Once the rotor starts moving at synchronous speed, this external source is shut off. In synchronous motor, the magnetic field of rotor is produced not by the magnetic field of rotor through induction. Instead, the magnetic field is generated by direct current supply. Therefore, the air gap between rotor and stator is not kept very small as in the case of induction motors.

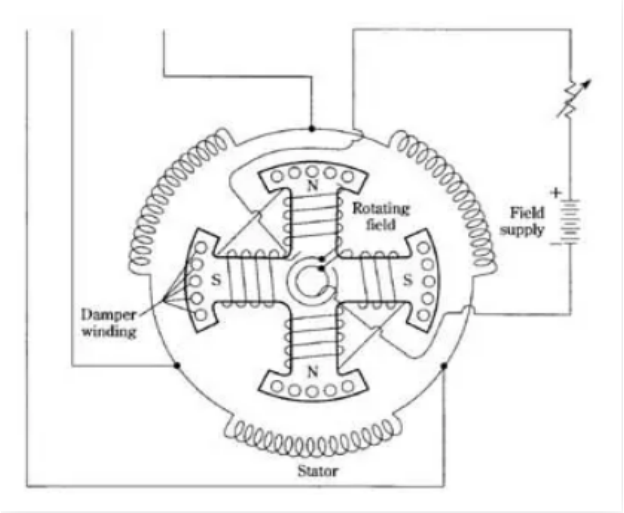


Figure 2:Working of synchronous motor

Starting Methods of synchrounous motor:

A synchronous motor is not inherently self-starting. Therefore, it requires some auxiliary means of starting. In order to start a synchronous motor, there are following two methods −

- Starting with an external prime mover
- Starting with damper windings

- ### SYNCHRONOUS MOTOR STARTING WITH AN EXTERNAL PRIME MOVER

In this method of starting a synchronous motor, an external motor is used to drive the synchronous motor. The external motor brings the synchronous motor to synchronous speed and then the synchronous motor is synchronized with the AC supply as a synchronous generator. Then the prime mover (i.e., the external motor) is disconnected. Once synchronized, the synchronous machine will operate as a motor.

Now, the mechanical load can be connected to the shaft of the synchronous motor. Since the load is not connected to the synchronous motor before synchronizing, thus the prime mover motor has to overcome only the inertia of the synchronous motor at no-load. Consequently, the rating of the prime mover or starting motor is much smaller than the rating of the synchronous motor.

Synchronous Motor Starting with Damper Windings

At present, damper windings are the most widely used method of starting a synchronous motor. A damper winding is made up of heavy copper bars inserted in slots cut into the pole faces of the rotor. These copper bars are then short-circuited by end rings at both ends of the rotor. Hence, this arrangement of copper bars and end rings form a squirrel cage winding. When the armature of the synchronous motor is connected to a three-phase supply, the synchronous motor with damper winding will start as a 3-phase squirrel cage induction motor. When the motor attains a speed nearer to the synchronous speed, the DC field excitation is applied to the rotor field windings. Then, the rotor will pull into step with the rotation magnetic field of the armature and hence the motor runs at synchronous speed. Some terms related to synchronous motor:

Synchronous Speed

$$N_s = \frac{120\ f_e}{P}$$

Where f = frequency of AC supply p= Number of poles

The speed is synchronous speed. It does not depend on load conditions.

Starting Torque: It is the torque (or turning effort) developed by the motor when full voltage is applied to its stator (armature) winding. It is also sometimes called breakaway torque.

Pull-in Torque: A synchronous motor is started as induction motor till it runs 2 to 5% below the synchronous speed. Afterwards, excitation is switched on and the rotor pulls into step with the synchronously- rotating stator field. The amount of torque at which the motor will pull into step is called the pull-in torque.

Pull-out Torque: The maximum torque which the motor can develop without pulling out of step or synchronism is called the pull-out torque.

3.Design:

3.1 PROBLEM FORMULATION:

Turning a 3-phase induction motor into a synchronous motor is a complex engineering project that involves several steps and considerations. To formulate the problem for this project, we can break it down into key components, including objectives, constraints, and the methodology we plan to use. Here's a structured problem formulation.

3.1.1. IDENTIFICATION OF SCOPE:

The project will focus on the conversion process itself, assuming the availability of an existing 3-phase induction motor.

3.1.2. FORMULATION OF PROBLEMS

Our main objective is to convert an existing 3-phase induction motor into a synchronous motor while achieving synchronous operation by maintaining constant speed as well as optimizing the motor's efficiency, power factor, and performance. The motor's power rating and application will be specified.

3.1.3 ANALYSIS:

- **Constraints and limitations:**
  Before diving down to the work, we needed to consider some constraints and limitations such as Budget constraints which limit the cost of the conversion project. The detailed cost will be attached to the report in the later section.
- **Modifications:**
  We needed to do two main modifications, one was the rewinding of the stator which was done for the purpose of learning and understanding the work principle of the stator. Secondly, we had to drill the rotor into four symmetric sections for the addition of another winding in order to provide the motor with a DC field excitation.
- **Risk Assessment:**
  We had to identify potential risks and develop mitigation strategies. While drilling the rotor there was a possibility of the laminated sheets of the rotor falling out, which might have ruined the entire mechanism of the rotor.

- **Cost Estimation:**

  We had to estimate the overall project cost, including materials, labor, and equipment.

3.2 DESIGN METHOD (PO(A)):

Coil pitch:

The distance between the two sides of an individual coil of an AC armature winding is termed the coil pitch. It is expressed as mechanical angle θ_m or electrical angle ρ

For the motor used in this project, the winding diagram is :

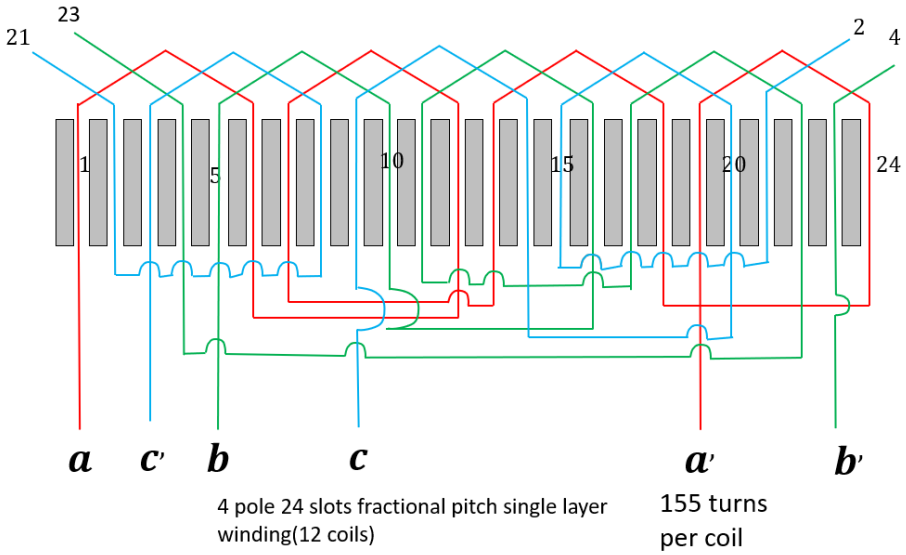


Figure 3: Winding diagram of the motor used

Hence, $\theta_m = \frac{90}{6} * 5 = 75°$ in mechanical degree or

$\rho = 150°$ *in electrical degree*

Pitch of the fractional pitch coil $= \frac{75°}{90°} = \frac{5}{6}$

Pitch factor, $k_p = \sin(\frac{\rho}{2}) = \sin\left(\frac{150}{2}\right) = \sin(75) = 0.9659$

Distribution factor, $k_d = \frac{\sin\left(\frac{n\gamma}{2}\right)}{n\sin\left(\frac{\gamma}{2}\right)} = \frac{\sin(2*30/2)}{2\sin\left(\frac{30}{2}\right)} = .9659$

Winding factor, $k_w = k_p * k_d = 0.9329$

Turns calculation:

We know that $E = \sqrt{2}\pi N f \varphi K_w$

Generally, a 1 HP 3-phase induction motor has an air gap flux density of (0.35-0.55) Wb/m^2. For the sake of simplicity, we can assume that the air gap flux, $B_{av} = 0.38$

Core diameter, D = 6 cm

Core length, L = 5.5 cm

Total flux in air gap, $\varphi = B_{av}A = 9.84 * 10^{-4}\ T$

Hence using the values found, $t = \frac{415/\sqrt{3}}{\sqrt{2}\pi * 50 * 9.84 * 10^{-4} * 0.9329} = 1173.77\ \ turns/phase$

$$\therefore Turns\ per\ coil\ is\ \frac{1173}{4} \cong 293\ turns\ per\ phase$$

Rated Current Calculation:

Rated line voltage = 415V (The range is 380-440V)

Rated Power Rating = 300W

Armature Rated Current Calculation:

$$I_{rated} = \frac{P}{\sqrt{3}V * cos\theta} = \frac{300}{\sqrt{3} * 415 * .6} = 0.6956\ A$$

This calculation is done on a per phase basis.

Wire size Calculation:

Cross sectional area of stator conductor, $A = \frac{I}{\delta} = \frac{0.6956A}{7.5\ A/mm^2} = .09275mm^2 = \mathbf{29SWG}$

3.3 CIRCUIT DIAGRAM:

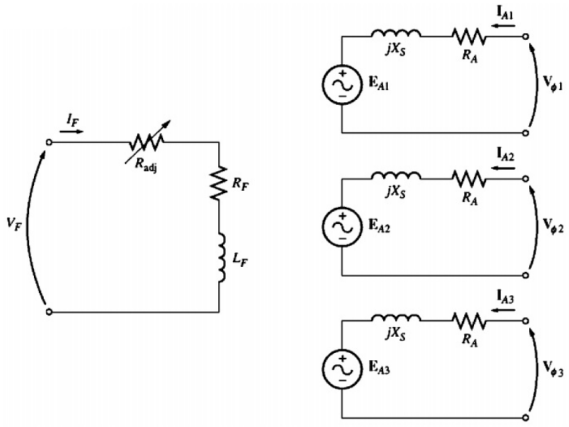


Figure 4: Circuit diagram of a 3-phase synchronous motor

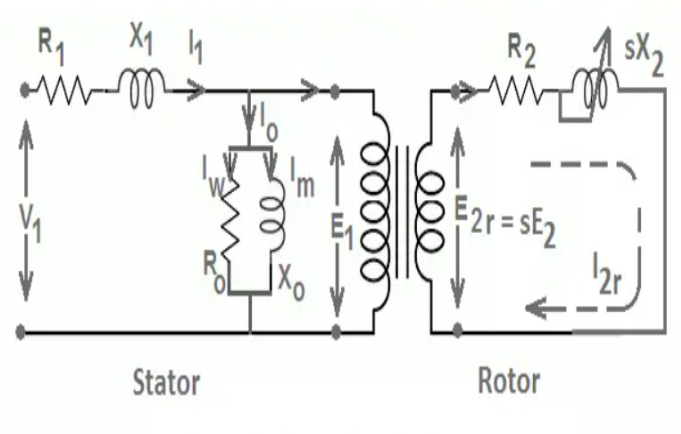


Figure 5 Circuit diagram of a 3-phase induction motor

3.4 SIMULATION MODEL OF A 3-PHASE SYNCHRONOUS MOTOR:

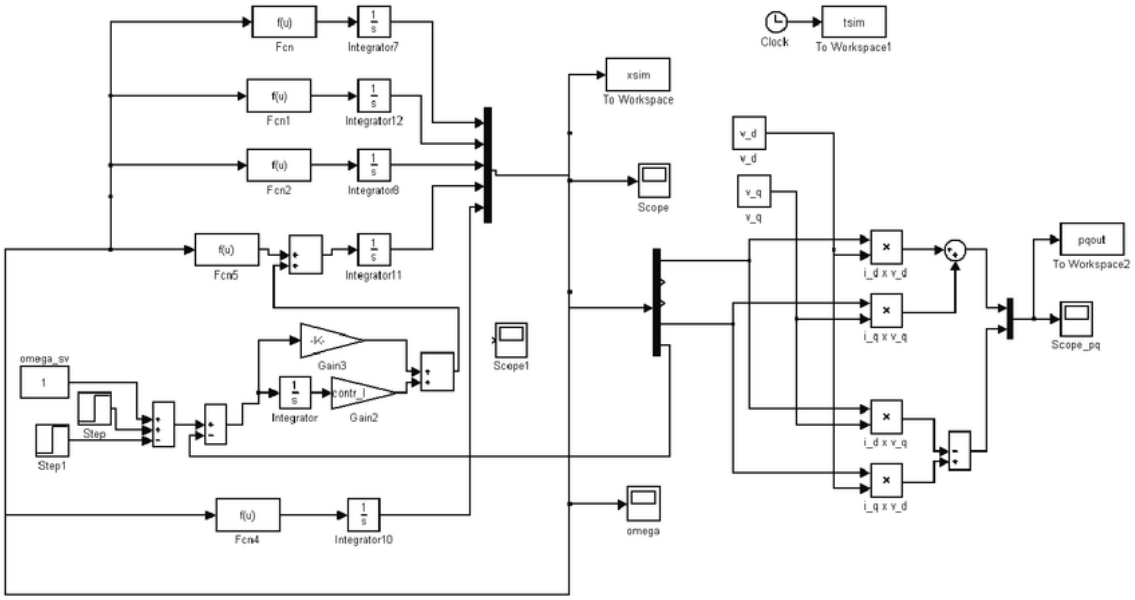


Figure 6: Simulation Model of a synchronous motor in MATLAB SIMULINK

4. Implementation:

Implementation of the theoretical knowledge, the calculation done for the design of the motor and tests performed to determine motor parameters achieved PO(d)

4.1: DESCRIPTION:

A three-phase induction motor and a synchronous motor have many things in common. In fact, conventional synchronous motors contain rotors with squirrel cage induction winding (the rotor that is used in squirrel cage induction motors). So we can think of synchronous motor as an induction motor with an external DC excitation into the rotor. This is why we started off our project by buying a Squirrel cage induction motor. The procedures that we took to transform the induction motor into a working synchronous motor is shown below,

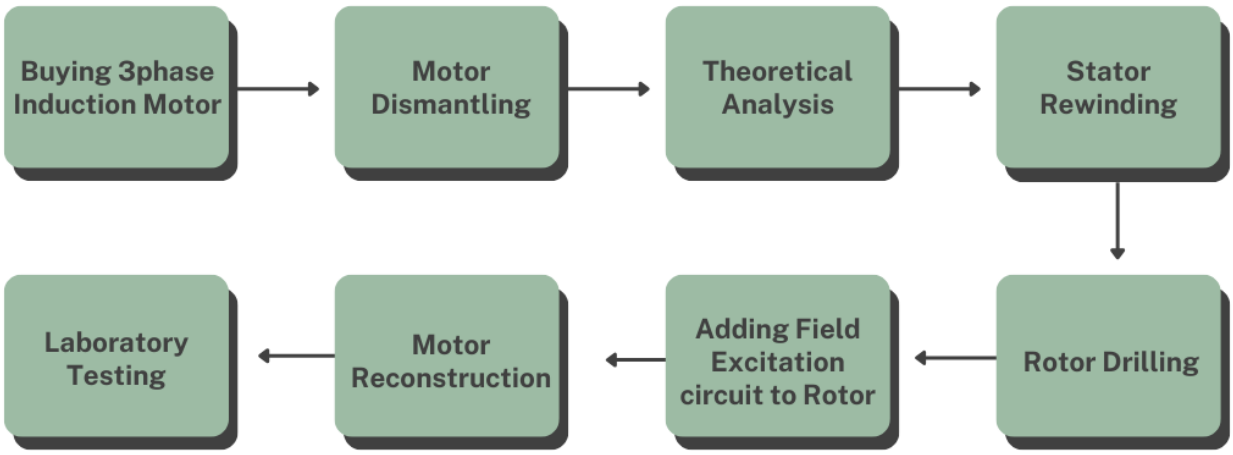


Figure 7: Workflow of the project

Step 1

A three-phase induction motor was bought with the intention of converting it into a synchronous motor. The specification of the motor is given below,

Type	Y2-80M2-4
Power (KW)	0.746 (1 hp)
Rated Speed	1395 rpm
Connection	Y
Phase	3
Rated Voltage	380-440
Rated Current	2.03 A
Normal Frequency	50 Hz
Insulation Type	CL F



Step 2

The motor was then dismantled. There are a total of 24 slots in the stator, each of which is filled with a conductor. There are insulating sheets placed inside the stator slots to provide insulation to the coils from the metal body of the stator as a safety measure. The coils are placed among the slots according



to the winding diagram shown below. As the motor is a 4-pole machine there are 6 slots in the stator pole.

Step 3

Since we wanted a different power rating than that of the initial motor, we cut the old stator windings and calculated the type of wire we need for both stator and rotor accordingly



Figure 8: 29 SWG and 31 SWG wire

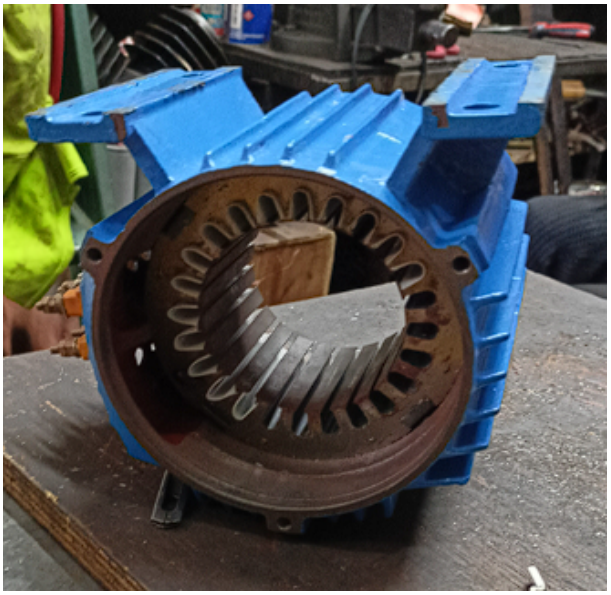
https://en.wikibooks.org/wiki/Engineering_Tables/Standard_Wire_Gauge

We selected 29 SWG.

Step 4

After buying the suitable wire, we moved our motor to the workshop and began the practical work required for our project. The first step was to carefully place insulation sheets in all the stator slots to prevent any of the stator coils from coming into contact with the core. Then we wound 12 coils of the wire we bought previously and placed them into the slots. Each coil consisted of 290 turns.

After placing the coils, we established the 3-phase Y connection again. This concluded our work on the stator.





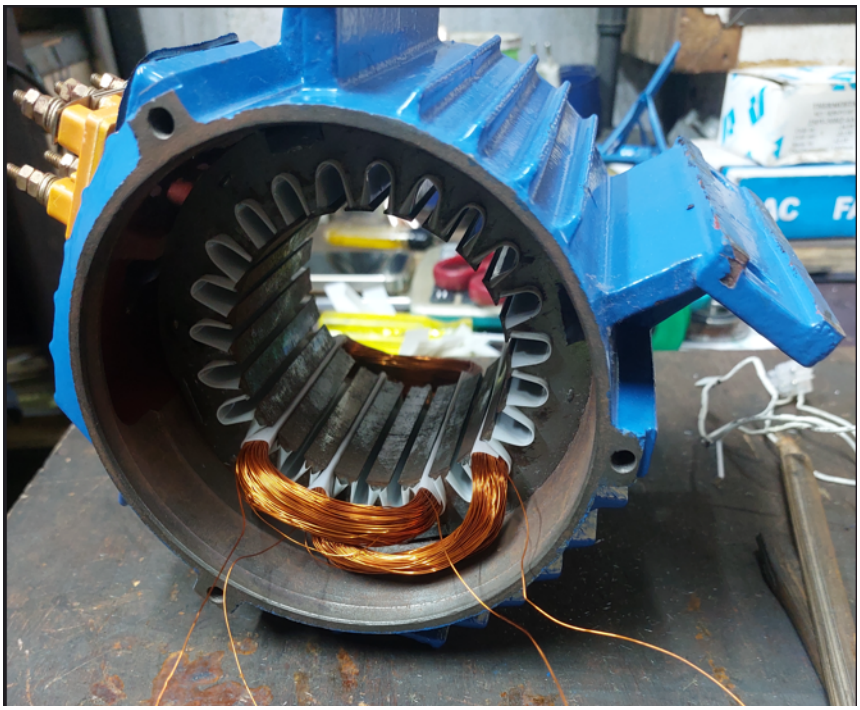


Figure 9: Stator rewinding

Step 5

Since we had to transform the induction motor to a synchronous motor, we needed to establish a DC field connection to the rotor. To make room for the field windings we drilled 4 holes (about 90 degrees apart) of approximately 10mm diameter.



Figure 10: Drilled Rotor

Step 6

After Drilling the rotor, we wound 2 coils through the 4 holes. 2 ends of the coils were brought out and connected to 2 leads. We mounted slip rings and brushes to provide the rotor with DC supply.



Figure 11: Rotor with DC field coil

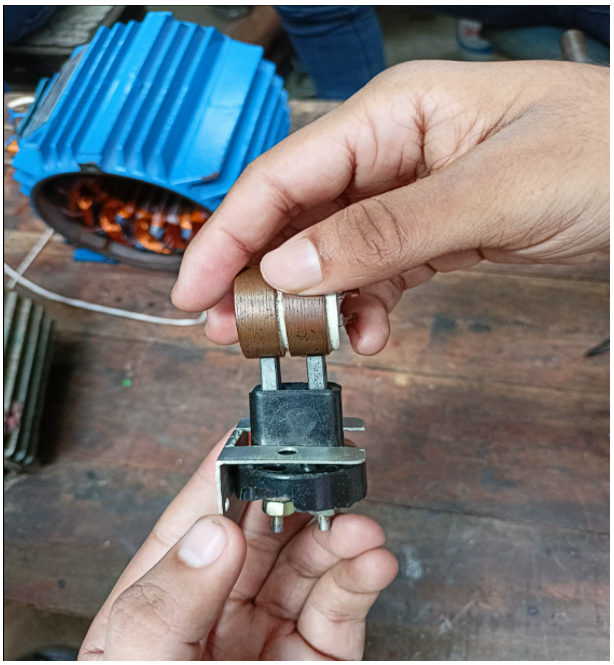


Figure 12: Carbon brush and slip ring

Step 7:

After Adding DC excitation circuit to the rotor, we reassembled the motor then took it to the Advanced machine lab for testing. It also provides current to the rotor of synchronous machine through slip rings. The exciter can be driven by a motor or shaft of a generator. It can be either self-excited or separately excited.

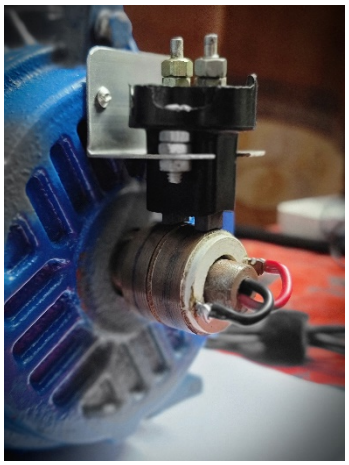


Figure 13: Slip ring and carbon brush

Step 8

In the lab, we did a test run and the motor seemed to work perfectly. Then with the help of Debashish Sir, we conducted no load and locked rotor tests on the motor. We also observed values for armature current for different field current.



Figure 14: Testing the motor

- **4.2 EXPERIMENT AND DATA COLLECTION:**

DC test:

Instead of applying DC voltage and measuring the DC current to determine the DC armature resistance, we measured the resistance directly from the multimeter.



Figure 15: Performing DC test with a multimeter

Field resistance, $R_f = 138.4\Omega$

Stator resistance:

In between line a & b: $R_{ab} = 145\Omega$

In between line b & c: $R_{bc} = 144\Omega$

In between line a & c: $R_{ac} = 149\Omega$

Per phase resistance $R_A = \frac{145+144+149}{3\times2} = 73\ \Omega$

Locked Rotor Test:

$I_L = 0.5\ A$
$V_L = 220\ V$
$P_{in} = 50\ W$

So, $|Z| = \frac{\frac{V_L}{\sqrt{3}}}{I_L} = \frac{220}{\sqrt{3}\times0.5} = 254.034\Omega$

Impedance Angle corresponding to locked rotor test,

$$\theta = \cos^{-1}\left(\frac{3\times P_{in}}{\sqrt{3}V_L I_L}\right) = \cos^{-1}\left(\frac{3\times 50}{\sqrt{3}\times 220\times 0.5}\right) = 38.066^\circ$$

Therefore, $\mathbf{R}_{syn} + \mathbf{R}_{rotor} = |Z|\cos\theta = 254.034\times\cos(38.066^\circ) = 200\Omega$

And $X_s = |Z|\sin\ \theta = 254.034 \times \sin(38.066°) = 156.62\Omega$

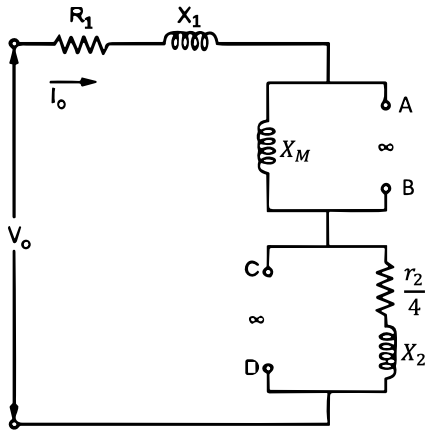
$$R_1 = 73\Omega$$
$$x_1 + x_2 = 156.62$$

Assuming design *B*

$$x_1 = 0.4 \times 156.62 = 62.648 \qquad x_2 = 0.6 \times 156.62 = 93.97$$

No Load Test:

$I_a = 0.2A$ $\eta = 1505\text{rpm}$ $P_{in} = 18 \times 3 = 54W$ $V_L = 350V$



At no load the power is dissipated as copper loss and rotational loss which includes mechanical, friction, windage and stray losses.

So, $P_{rot} = P_{in} - I^2R_A = 54 - 0.2^2 \times 73 = 51W$

$$x_1 + x_m = \frac{350}{\sqrt{3} \times 0.2} = 1010.36$$
$$x_m = 947.717$$
$$R_2 = 200 - 73 = 127\Omega$$

I_A vs I_F(V curve):

I_F(A)	0	0.05	0.1	0.2	0.25	0.3	0.35	0.4	0.5	0.55	0.6	0.7	0.75
I_A(A)	0.2	0.17	0.15	0.1	0.08	0.06	0.06	0.06	0.1	0.12	0.14	0.19	0.21

The obtained data is used to fit a second-order polynomial by polynomial regression since the armature current vs. field current of a synchronous motor is a " V " curve and then, plotted in Python.

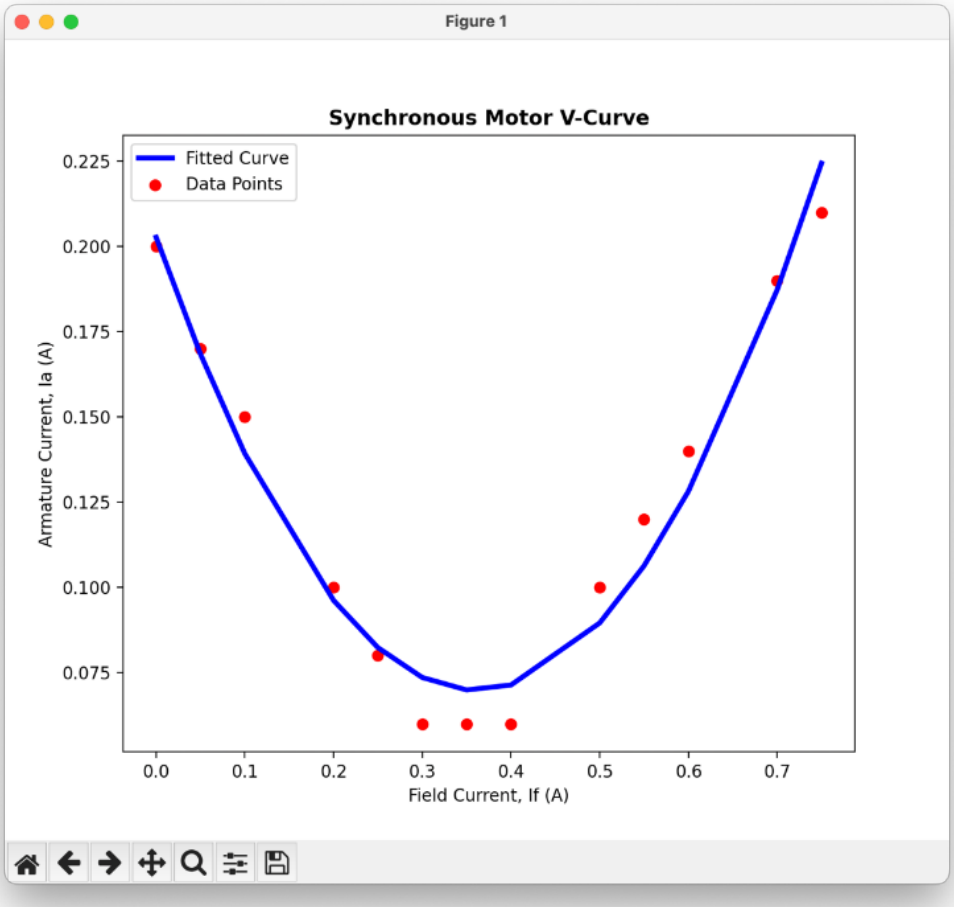


Figure 16:V curve using python

Python Code to Plot This Graph:

```
import numpy as np
import matplotlib.pyplot as plt

# Given data points
if_data = np.array([0, 0.05, 0.1, 0.2, 0.25, 0.3, 0.35, 0.4, 0.5, 0.55, 0.6, 0.7, 0.75])
ia_data = np.array([0.2, 0.17, 0.15, 0.1, 0.08, 0.06, 0.06, 0.06, 0.1, 0.12, 0.14, 0.19, 0.21])

# Fitting the curve from the data
coeff = np.polyfit(if_data, ia_data, 2)

# Generate fitted data points
fitted_ia_data = np.polyval(coeff, if_data)

# Plotting
plt.figure()
plt.plot(if_data, fitted_ia_data, 'b', linewidth=3, label='Fitted Curve')
plt.scatter(if_data, ia_data, c='r', label='Data Points')
plt.xlabel('Field Current, If (A)')
plt.ylabel('Armature Current, Ia (A)')
plt.title('Synchronous Motor V-Curve', fontweight='bold')
plt.legend()
plt.show()
```

5. Design Analysis and Evaluation

5.1 Novelty:

In synchronous motor design novelty refers to innovative and unique aspects or features that distinguish the motor from existing designs. We bought a three-phase induction motor and reconstruct it into a synchronous motor. There was no new design in our synchronous motor, but we design the existing rotor of induction motor into dc excitation field rotor of synchronous motor.

5.2 Design Considerations:

To ensure the safe and efficient operation of synchronous motors, it's important to take certain precautions and follow best practices. Here are some precautions and maintenance steps to consider:

5.2.1 Considerations to Public Health And Safety:

In motor design, safety is paramount.

Electrical Hazards:

Induction motor rewinding involves working with high-voltage electrical circuits. All equipment must be properly de-energized and locked out before any rewinding work begins. This includes verifying the absence of voltage using appropriate testing tools.

Personal safety awareness:

One should wear suitable insulated gloves, safety goggles, flame-resistant clothing, and protective footwear. These measures protect against electrical shock, burns, and other potential hazards.

Ventilation: maintain proper ventilation in the workspace is essential, especially when using insulating materials and adhesives. Proper ventilation helps dissipate fumes and prevents the buildup of harmful gases that could be released during the rewinding process.

Fire Safety:

Rewinding may involve the use of materials like insulating varnishes and tapes that are flammable. Maintain fire safety precautions, such as the presence of fire extinguishers and a clear evacuation plan in place to mitigate the risk of fire.

Work Area Organization:

We ensured the work area is clean and well-organized. Cluttered workspaces can increase the risk of accidents and make it difficult to respond quickly to emergencies.

Machine Guarding:

We Ensured that all rotating parts of the motor, such as the rotor and moving components, are properly guarded. This prevents accidental contact and injuries during the rewinding process.

Chemical Handling:

When using chemicals like solvents or cleaning agents, we followed proper handling procedures, and store them in designated areas. aware of the hazards associated with these chemicals and used them in well-ventilated spaces.

5.2.2 Considerations to Environment:

Energy Efficiency: Efficient energy usage is a key environmental consideration. Synchronous motors ensure maximum energy efficiency, reducing power consumption and minimizing waste heat production. This not only reduces operating costs but also decreases the environmental impact by reducing the need for additional power generation.

Emissions Reduction: Motors should be designed to produce minimal emissions, especially in the case of combustion engines. This involves optimizing combustion processes, using clean fuels, and incorporating emission control technologies.

Environmental Materials: One should use environmentally friendly materials in motor construction. This includes choosing materials that are recyclable, biodegradable, or have a lower environmental footprint throughout their lifecycle.

5.2.3 Considerations to Cultural and Societal Needs:

Local Compatibility: Motors designed for specific regions should be compatible with the local electrical grid and voltage standards. This ensures that the motor can be easily integrated into the existing infrastructure without the need for costly modifications.

Affordability: Consider the economic conditions of the target market. The motor's design should aim to be cost-effective and affordable for the intended users. This may involve optimizing manufacturing processes and materials to reduce production costs.

Cultural Sensitivity: In some cases, motors may be used in culturally or historically significant areas. Design should consider any cultural or historical sensitivities, such as the preservation of heritage sites.

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5.3 Investigations:

Literature Review: we conduct a comprehensive literature review to understand existing motor designs, emerging technologies, and relevant research.

Data Analysis and Interpretation:

Efficiency Calculation (per phase):

We measured the rotational loss, $P_{rot} = 51.08W$

Copper loss at rated armature current, $P_{cu} = 35.32\ W$

So, minimum efficiency, $\eta_{min} = \frac{100-51.08-35.32}{100} \times 100\% = 13.6\%$

The rated armature current in this instance was measured at power factor 0.6. Therefore, the copper loss calculated here is at its greatest, and the related calculated efficiency is at its lowest possible level.

However, as can be seen from the V curve, the armature current can be reduced by altering the synchronous motor's field current. In this scenario, efficiency will be at its highest and copper loss will be at its lowest.

Then, $I_A = \frac{100}{\frac{415}{\sqrt{3}}} = 0.4174A$

So, copper loss in this case, $P_{cu} = 12.72W$

So, theoretical maximum efficiency, $\eta_{max} = \frac{100-51.08-12.72}{100} \times 100\% = 36.2\%$

So, the motor efficiency range is 13.6% to 36.2%.

5.4 LIMITATIONS OF TOOLS:

1. Drilling of rotor was very challenging. As we don't have drilling machine, we have to drill it from workshop.

2. For rewinding wire in stator and rotor we use a handmade winding machine.

5.5 IMPACT ASSESSMENT:

Societal and cultural issues: The design does not have any negative impacts on the target culture or society. designing a synchronous motor with consideration for public health and safety, environmental

impact, and cultural and societal needs is essential to create a motor that not only functions efficiently but also align with ethical and sustainable principles.

Health and safety issues: The design does not pose any risks to public health or safety. By taking the necessary precautions we help to protect ourselves from the health risks associated with working on synchronous induction motors.

Legal issues: Ensure compliance with local, state, and national electrical and workplace safety regulations, securing the required permits and licenses for the motor rewinding project. Properly handle and dispose of hazardous materials, such as old insulation materials and chemicals, in accordance with environmental regulations to prevent potential legal issues.

5.6 Sustainability and Environmental Impact Evaluation:

In the pursuit of sustainable practices, this evaluation encompasses various facets, including the motor's energy efficiency, resource utilization, and environmental footprint. We minimize energy consumption, reduce material waste, and employ environmentally friendly material.

5.7 Ethical Issues:

By acknowledging and adhering to ethical principles, we uphold the values of professionalism and integrity in the field of induction motor rewinding, ensuring both ethical and technical excellence.

6. Reflection of Individual Teamwork:

6.1. Individual Contribution of Each Member:

Buying the induction motor:

2006099, 2006125, 2006129

Disassembling motor:

2006100, 2006101, 2006103, 2006125, 2006129

Cutting the winding:

2006099,2006101, 200125, 2006127, 2006129

Calculating the Winding of the Rotor and Stator:

All the members of the group

Drilling the Rotor:

For drilling the rotor, we took the help of a local engineering workshop. But all the members of the group were present there.

Buying wire for stator and rotor:

2006125, 2006127, 2006101

Inserting insulation paper:

2006099,2006101, 2006102, 2006104, 2006125, 2006126, 2006128, 2006130, 2006129

Rewinding the rotor:

All the members of the group.

Rewinding the Stator:

All the members of the group.

Setting Up Slip Ring and Brushes:

2006100, 2006102, 2006103, 2006125, 2006126, 2006127, 2006128, 2006129, 2006130

No Load Test:

2006100, 2006102, 2006103, 2006104, 2006125, 2006126, 2006127, 2006128, 2006130

Block Rotor Test:

2006100, 2006102, 2006103, 2006104, 2006125, 2006126, 2006127, 2006128, 2 006130

Checking Field Current and Corresponding Armature Current to Find V curve:

2006100, 2006102, 2006103, 2006104, 2006125, 2006126, 2006127, 2006128, 2006130

- ## 6.2 MODE OF TEAMWORK:

Our team was committed towards the common goal of making a synchronous motor. We worked together to complete it within the given time limit. We shared ideas about the procedure for the project. Then we created a flow chart to complete the project. After that, we started according to our plan. In some cases, we were getting stuck. Then we discussed and tried to understand our own points about the adjustments we should make to go further.

- ## 6.3 DIVERSITY STATEMENT OF TEAM:

This is the first Hardware project we have done. In this project work, we two groups have worked together. Every member of group showed his/her individual skills to make successful the project. Both theoretical and management skills had been combined. For any theory and practical work-related query, we consulted with our Lab instructor. We have been able to gain real engineering knowledge and learn how to work with various barriers in the practical field. We successfully completed our project with obtaining knowledge and skills.

- ## 6.4 LOGBOOK OF PROJECT IMPLEMENTATION:

Date	**Our Achievements**	**Individual Role**	**Group Role**	**Comments**
16/07/23	Buying an Induction motor	2006099, 2006125, 2006129		Money receipt added
17/07/23	Dissembling of motor	2006100, 2006101, 2006103 2006125, 2006129	Discussion on complete project	
18/07/23	Wire cutting from stator	2006099, 2006127 2006125,		Simple effort needed to do it

		2006129		
23/07/23	Consulting with Lab instructor	2006099, 2006100, 2006101, 2006103 2006125, 2006129		
29/07/23	Approximate and rough calculation.	All member		we took help from internet for proper assumption.
15/08/23	Rotor drilling	All member		We took help from Local engineering shop
16/08/23	Buying wire for stator and rotor	2006127, 2006125		
01/09/23	Stator winding With new wire	All member		
01/09/23	Inserting insulation paper	2006099, 2006101, 2006102, 2006104, 2006125, 2006126, 2006128, 2006129, 2006130		

03/09/23	Rotor rewinding	All member		
04/09/23	Addition of slip ring and brush Complete assemble of motor	2006100, 2006102, 2006104, 2006125, 2006126, 2006128, 2006129, 2006130		
09/09/23	Various test (No load test, Block rotor test, characteristics of V vs armature and field current)	2006100, 2006102, 2006104, 2006125, 2006126, 2006128, 2006129, 2006130		By the help of Lab instructor, we completed our test and got ensured about our machine
09/09/23	Preparation of Report	All member		
11/09/23	Project presentation	All member		

7. Communication:

- Ramit Dutta Sir,
  Lecturer, Department of EEE, BUET.
  who assigned the project to our team.

- Sir, Ashikur Rahman Jowel

Lecturer, Department of EEE, BUET.
who assigned the project to our team.

- Mr. Debasis Das,
  Senior Tech. Officer, Department of EEE, BUET.
  in order to get technical support and advice. He provided us with details on the project's cost and labor.

- Since synchronous motor of low power was not available in the market, we were advised to buy an induction motor. The induction motor ought to be converted into a synchronous motor. According to Debasis Sir, the stator should be wound exactly in the same manner as an induction motor and for supply dc power, the rotor should contain windings and so, it should be drilled to place the windings. He advised us that professionals should handle this as it's the most sensitive part of the whole work.
- We bought the induction motor from a machinery shop, Nawabganj. After examining the motor, we then turned over the motor to them. They started winding the stator using the wire specification suggested by us. We finished our job while working with regular shop employees.
- To drill the rotor, we contacted another workshop at Bakshi Bazar, suggested by Debasis Sir. They worked with the rotor for a day long, and we observed the whole working process and participated in rotor winding.
- Then we took our motor to the advanced machine lab for testing. Debashish Sir diligently examined the motor, and we wrote down the relevant data for motor's equivalent circuit.
  We tried to communicate effectively on complex engineering activities with the electrical and electronic engineering and other inter-disciplinary communities and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. Thus, we can expect PO(j) to have been achieved.

8. Project Management and Cost Analysis (PO(k))

The tabular form of our costs in different sectors of our project is given below:

Name of equipment/activity	Cost(BDT)
3 Phase Induction Motor	5500
29-gauge stator coil (1 kg), insulating paper, burnish	1950
Shaft, Slip Ring, brush	2500
31-gauge rotor coil	500
Rotor drilling	3000
Technical assistance	1000
Transportation	2000
Total	16450

Timeline Of Project Implementation:

Activity / Task	Starting Date	Duration(days)
Buying motor	16.07.23	1
Motor dismantling	17.07.23	1
Theoretical Analysis	29.07.23	4
Stator Rewinding	01.09.23	5
Rotor Drilling	15.08.23	7
Rotor Rewinding	03.09.23	3
Motor Reconstruction	05.09.23	1
Laboratory Testing	09.09.23	1
Report Finalization	09.09.23	2
Project submission	11.09.23	1

We tried to minimize the cost as much as possible, make decisions and finance unitedly.

9. Future Works

In this project, we tried to demonstrate how to operate a three-phase synchronous generator as a synchronous motor. It is obvious that our implementation of the foreign components is not aligned accurately with the device's internal spacing and that causes the device to be low efficient. To operate as a synchronous motor, the motor speed needs to be synchronized with supply frequency. We need to adjust the DC excitation to the rotor so that the motor locks into the synchronous speed of the system to which it is connected. Synchronization is essential for the motor to maintain a constant speed and phase relationship with the power grid. Another thing to be noted is the damper winding. Though we did not use the damper winding. The damper winding is typically used in synchronous motor operation and reduces oscillation. We need to make sure the damper winding is properly connected and in good condition. Damper windings are usually short-circuited loops on the rotor, and their design depends on the specific motor requirements.

Efficiency Improvements

1. **Component Alignment**: The first and most immediate area of focus will be to improve the alignment of foreign components with the device's internal spacing. This will help in reducing internal losses and improve the overall efficiency of the system.
2. **Material Optimization**: Exploring materials with better electromagnetic properties can significantly reduce the losses and improve efficiency.

Synchronization Techniques:

1. **Advanced Control Systems**: To facilitate easy and accurate synchronization with the grid, we plan to incorporate a microcontroller-based synchronization mechanism that will automate the adjustment of DC excitation to the rotor.
2. **Adaptive Feedback Systems**: Integrate an adaptive feedback system that adjusts rotor excitation in real-time, ensuring the motor speed always matches the grid frequency.

Damper Windings

1. **Inclusion and Testing**: Since damper windings play a significant role in reducing oscillations, our next phase will involve incorporating these into our design and rigorously testing their effectiveness.
2. **Custom Design**: Investigate the need for custom-designed damper windings to meet our motor's specific requirements.

Grid Compatibility

1. **Grid Integration Tests**: Conduct a series of tests to evaluate how well the motor integrates with different kinds of power grids, focusing especially on its capability to maintain a constant phase relationship.
2. **Power Quality Analysis**: Evaluate the impact of the synchronous motor on the power quality of the grid to which it is connected.

Safety and Reliability

1. **Failure Modes Analysis**: Perform a comprehensive failure modes and effects analysis (FMEA) to identify potential points of failure and improve reliability.
2. **Safety Measures**: Incorporate additional safety measures, like overload protection and emergency stop features, to ensure safe operation.

Scalability

1. **Scalable Designs**: Investigate the scalability of our design to higher or lower power ratings to cater to different application needs.

Documentation and User Training

1. **User Manuals**: Develop comprehensive user manuals and training materials to facilitate easy adoption of the technology.
2. **Online Support**: Establish a robust online support system to resolve issues and update users on best practices.

By tackling these areas in our future work, we aim to develop a more efficient and reliable synchronous motor system that is easily adaptable to varying grid conditions.

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